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CPS Attack and Defense Testbed

Semester 1

September 28

2026

Requirements

Functional

- Connect Allen-Bradley, Siemens, and Modicon PLCs
- Support EtherNet/IP, PROFINET, and Modbus TCP
- Read PLC and sensor data through the historian
- Run reusable Python test/attack scripts
- Record cyber and hardware attack activity
- Restore PLC and sensor operation after testing

Non-Functional

- Isolated laboratory network
- Historian updates PLC/sensor changes within 2 sec
- Store events within 5 sec
- Maintain repeatable test results
- Support PLC profiles for plug-and-play configuration
- Protect passwords and PLC credentials



Tool Selection

PLC Programming

- Allen-Bradley — Connected Components Workbench / Rockwell software
- Siemens — STEP 7 / TIA Portal
- Schneider Modicon — EcoStruxure / SoMachine Basic



Tool Selection

Testing & Networking

- Python
- Wireshark
- Scapy

Historian

- HTML
- CSS
- JavaScript
- Python
- Docker



Tool Selection

Hardware

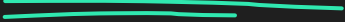
- Allen-Bradley PLC
- Siemens PLC
- Schneider Modicon PLC
- TP-Link / D-Link switches
- ESP32
- Sensors
- Trainer

Milestone 1 Completion Matrix

Task	Completion %	Karla	Zach	To do
1. Compare and Select Technical Tools	90%	50%	50%	Finish getting Siemens Step 7
2. “Hello World” demos	100%	50%	50%	None
3. Resolve Technical Challenges	100%	50%	50%	None
4. Design Document	100%	50%	50%	None
5. Requirements Document	100%	75%	25%	None
6. Test Document	100%	25%	75%	None

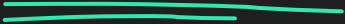
Test Plan

Test Items



- Allen-Bradley PLC
- Siemens PLC
- Schneider Modicon PLC
- PLC-to-network communication
- HMI and historian
- Sensors
- Network switches
- ESP32 interception module
- Python attack/test scripts
- Packet capture and logging

Protocols



- EtherNet/IP
- PROFINET
- Modbus TCP

Testing Levels

- Component Testing
- Component Integration Testing
- System Testing
- Demonstration / Acceptance Testing

Testing Scenarios

- Normal operation
- Connection loss & recovery
- Protocol misconfiguration
- Sensor interruption
- Attack-script repeatability
- PLC / historian communication

Pass Criteria

- Expected PLC behavior occurs
- Historian/network logs match event
- Test produces repeatable results
- System returns to baseline

Resolved Technical Challenges

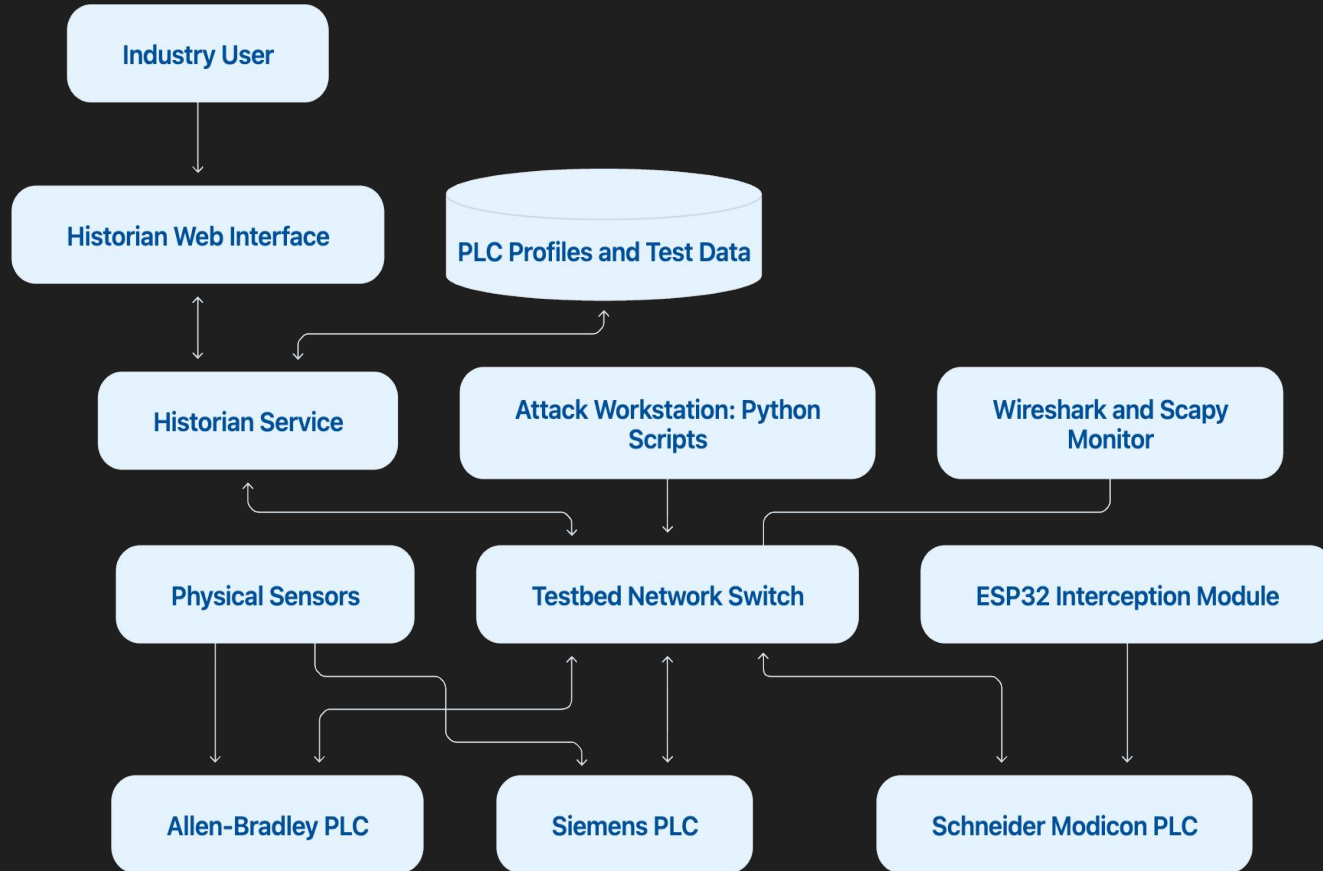
- Allen-Bradley software installed and configured
- Allen-Bradley PLC connected and programmed
- PLC/trainer I/O mapping documented
- Historian PC reflashed and Ethernet drivers restored
- TP-Link switch configured
- Power distribution created using DIN rail terminal blocks

Remaining Technical Challenges

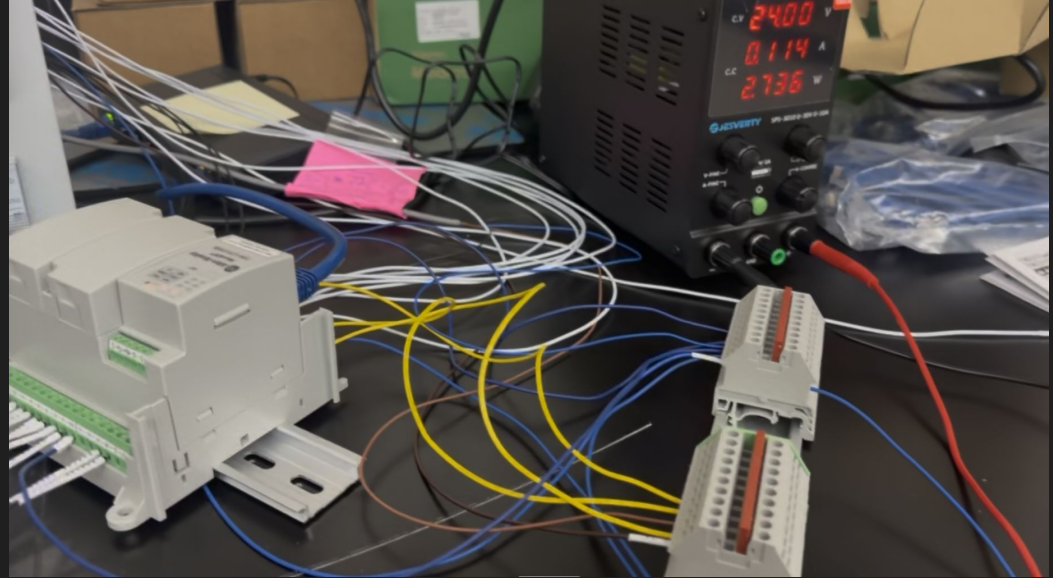
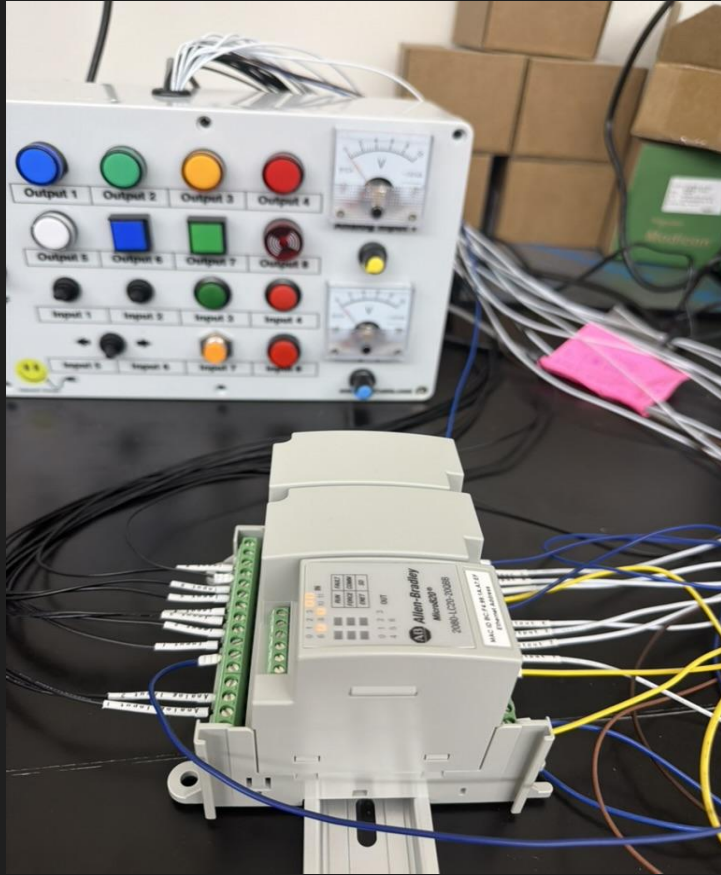
- Obtain Siemens STEP 7 access



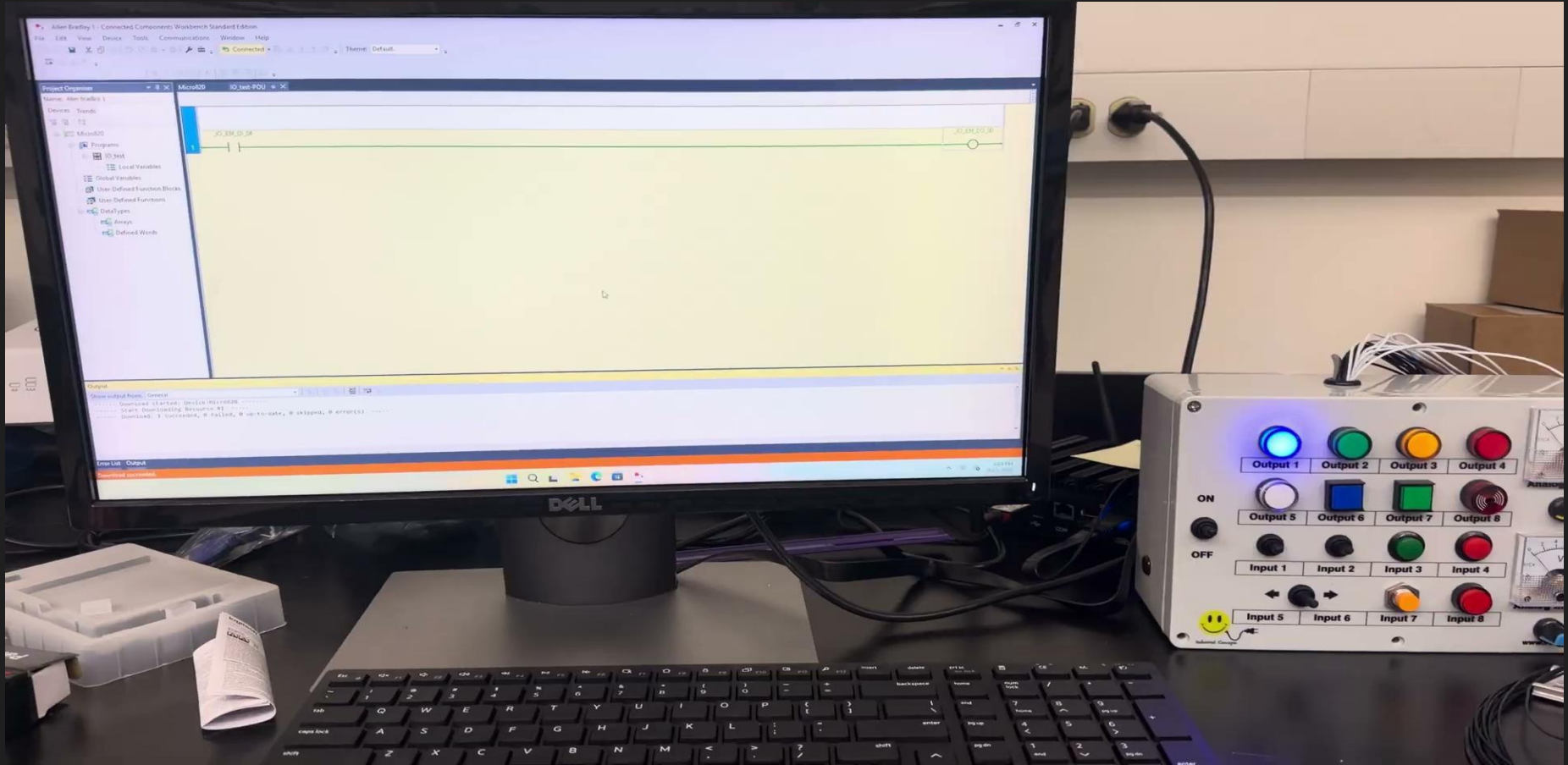
System Overview



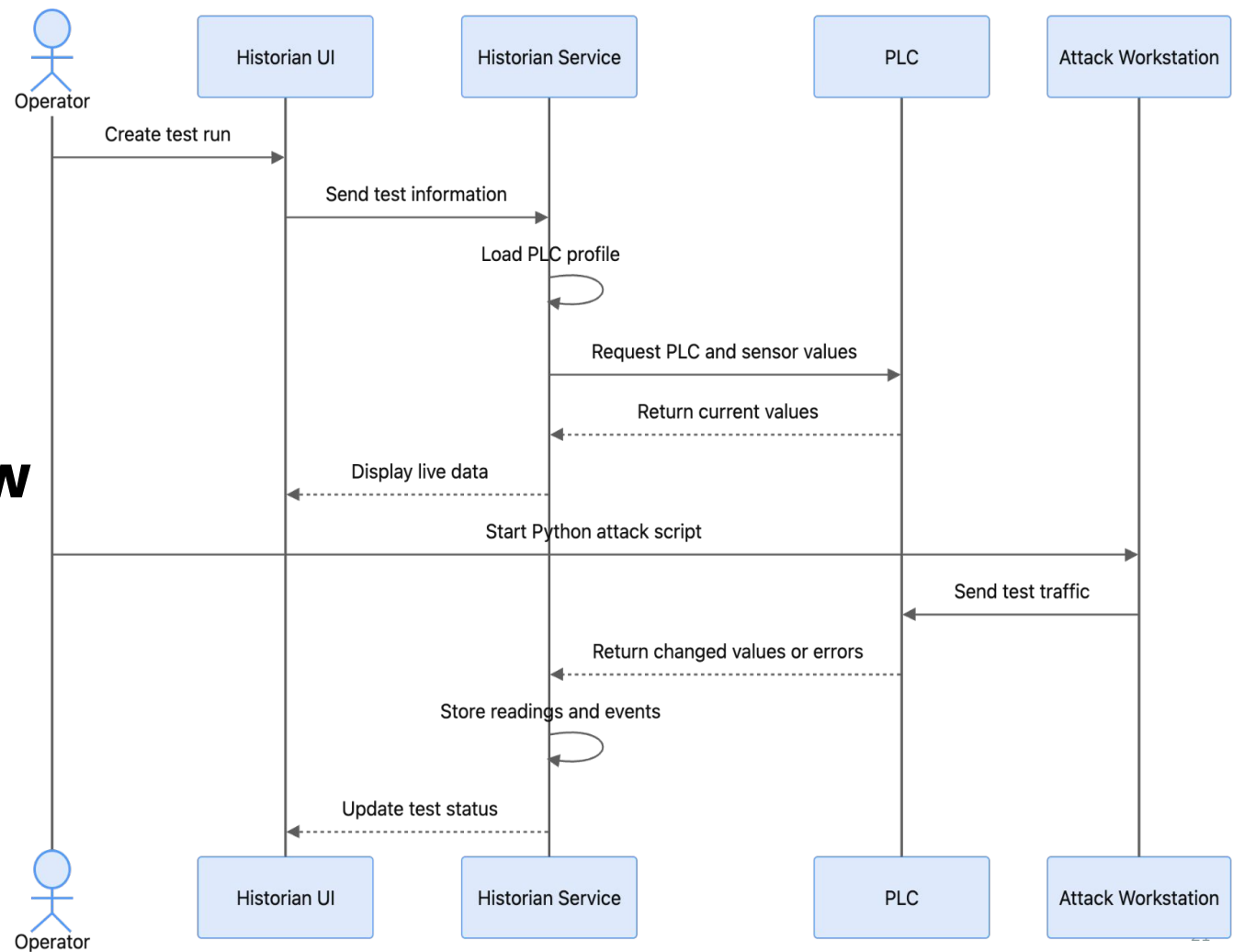
Milestone 1 Setup



Allen Bradley PLC "Hello World" Demo



Design Overview



Milestone 2 Task Matrix

Task	Karla	Zach
1. Demo internal network attacks	Set up PLCs so that when under attack, we can observe the physical outcome.	Reflash the second computer & make it able to see the network PLCs.
2. Set up sensor connection to all 3 PLCs	Set up the sensors that the PLCs will connect to. Make sure these also affect the PLC trainer readout.	Allow the historian to see all sensor data & create a logging system for it.
3. Finish setting up the Siemens PLC	Add Siemens PLC to the current hardware testbed setup, including software.	Acquire & setup Siemens Step 7 to interface with the PLCs through the switch.



Questions?